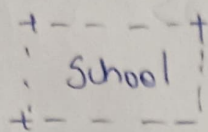


# Self Problem

Problem: School & Students with Courses Association and Aggregation

## Class Diagram



-name: String

+students: List<Student>

+-----+

+addStudent(student: Student): void

+removeStudent(student: Student): void

+-----+



+-----+

| Student |

+-----+

-name: String

-id: String

-courses: List<Course>

+-----+

+enroll(course: Course): void

+viewCourses(): List<Course>

+-----+

↔ Association (many to many)

+-----+

| Course |

+-----+

-title: String

-code: String

-student: List<Student>

+-----+

+addStudent(student: Student): void

+viewStudents(): List<Student>

+-----+

Reet Bhardwaj

Sec K

15



## Problem 2: University with Faculties and Department (composition and Aggregation)

Reet Bhardway

See k

Roll no 15

+-----+  
| University |

+-----+

| - name: String

| - department: List < Department > |

| - faculty: List < faculty > |

+-----+

| + add department (dep : Department) : void |

| + add faculty (faculty : faculty) : void |

+-----+

□ Composition



+-----+

| Department |

+-----+

| - name: String |

+-----+

◇ Aggregation



+-----+

| Faculty |

+-----+

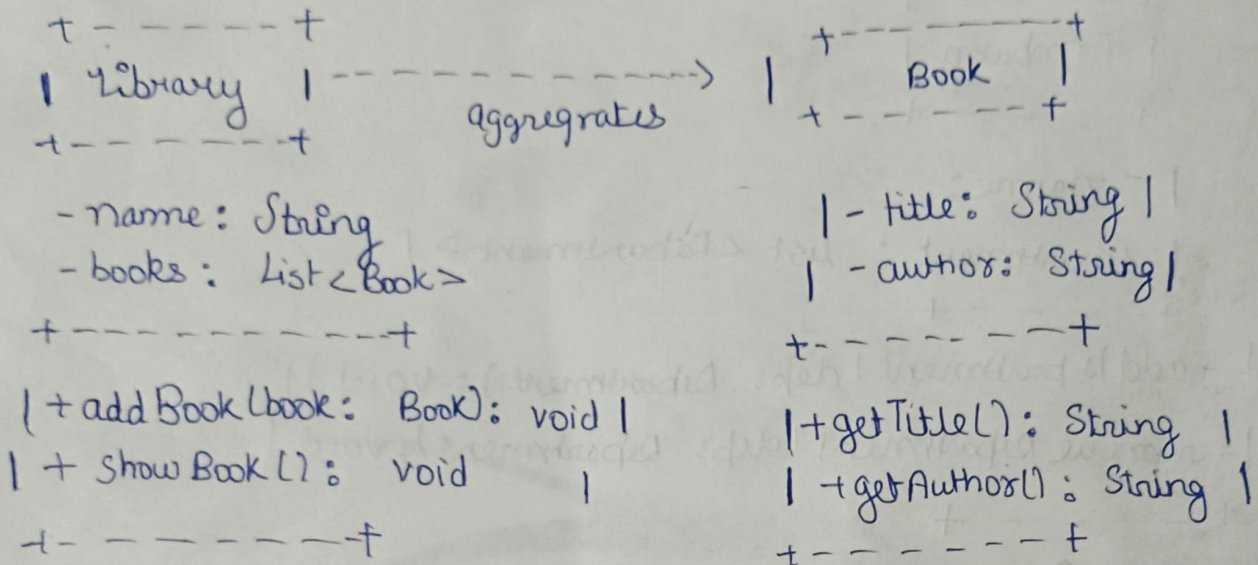
| - name: String |

| - title: String |

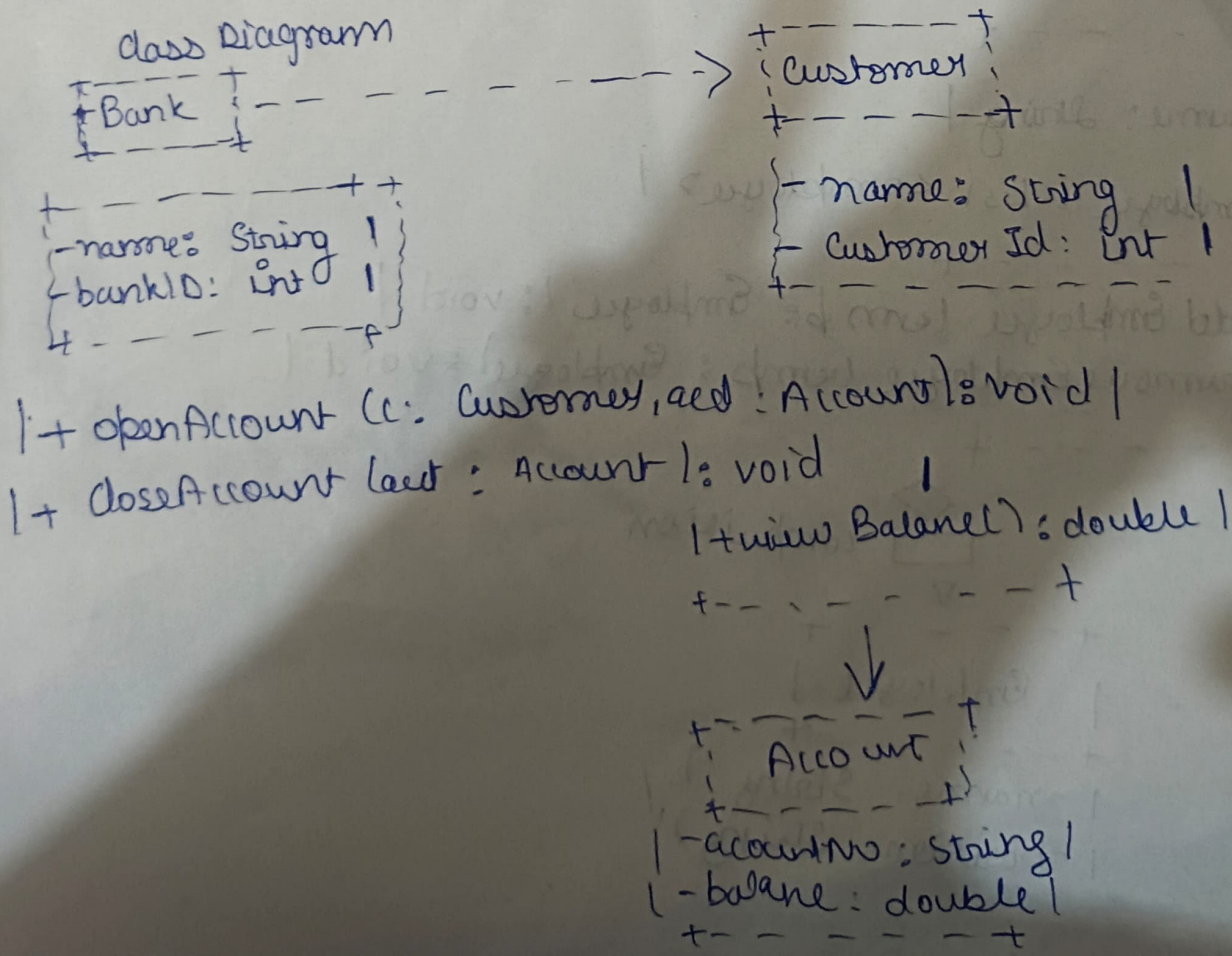
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## Problem 1 Library and Book (Aggregation)



## Problem 2 Bank and Account Holder (Associate)



## Problem 2: University with Faculties and Department (Composition and Aggregation)

Reet Bhardway

See k

Roll no 15

+ - - - - +

| University |

+ - - - - +

- name: String

- department: List < Department > |

- faculty: List < faculty > |

+ - - - - +

+ add Department (dep: Department): void

+ add Faculty (faculty: Faculty): void

+ - - - - +

□ Composition



+ - - - - +

| Department |

+ - - - - +

- name: String |

+ - - - - +

◇ Aggregation



+ - - - - +

| Faculty |

+ - - - - +

- name: String |

- title: String |

+ - - - - +



# Problem 3 Company & Departments Composition

Rec Branching

Class Diagram

See K  
15

```
+-----+
| Company |
+-----+
```

| - name:

| - department: List <Department> |

```
+-----+
```

| + add Department (dep: Department): void |

| + remove Department (dep: Department): void |

```
+-----+
```



composition



```
+-----+
| Department |
+-----+
```

```
+-----+
```

| - name: String |

| - employee: List <Employee> |

```
+-----+
```

| + add Employee (emp: Employee): void |

| + remove Employee (emp: Employee): void |

```
+-----+
```



composition



```
+-----+
```

| Employee |

```
+-----+
```

| - name: String |

| - position: String |